

Chengpiao Huang

chengpiao.huang@columbia.edu | <https://ch3702.github.io/>

Education

Columbia University

PhD student, Industrial Engineering and Operations Research

Advisor: Kaizheng Wang

New York, United States

September 2022 – Present

The Chinese University of Hong Kong, Shenzhen

Bachelor of Science, Mathematics and Applied Mathematics

Shenzhen, China

September 2018 – May 2022

Research Papers

The default author order is alphabetical.

- **LLM-Powered Virtual Population for Demand Simulation and Pricing**
Chengpiao Huang, Kaizheng Wang
Preprint, 2026.
- **SYN-DIGITS: A Synthetic Control Framework for Calibrated Digital Twin Simulation**
Grace Jiarui Fan, Chengpiao Huang, Tianyi Peng, Kaizheng Wang, Yuhang Wu
Preprint, 2026.
- **The Nonstationarity-Complexity Tradeoff in Return Prediction**
Agostino Capponi, Chengpiao Huang, J. Antonio Sidaoui, Kaizheng Wang, Jiacheng Zou
Preprint, 2025.
- **How Many Human Survey Respondents is a Large Language Model Worth? An Uncertainty Quantification Perspective**
Chengpiao Huang, Yuhang Wu, Kaizheng Wang
Preliminary version appeared in *International Conference on Machine Learning (ICML)*, 2025.
- **Distribution-Free Predictive Inference under Unknown Temporal Drift**
Elise Han, Chengpiao Huang, Kaizheng Wang
Preprint, 2024.
- **Model Assessment and Selection under Temporal Distribution Shift**
Elise Han, Chengpiao Huang, Kaizheng Wang
International Conference on Machine Learning (ICML), 2024.
- **A Stability Principle for Learning under Non-Stationarity**
Chengpiao Huang, Kaizheng Wang
Operations Research 73 (6), 3044-3064, 2025.
- **Relaxation-Free Min- k -Partition for PCI Assignment in 5G Networks**
Yeqing Qiu, Chengpiao Huang, Ye Xue, Zhipeng Jiang, Qingjiang Shi, Dong Zhang, Zhi-Quan Luo
IEEE Transactions on Signal Processing 73, 3931-3946, 2025.

Honors and Awards

- Outstanding Student Service Award in IEOR, Columbia University 2026
- Deming Doctoral Fellowship, Columbia University 2025–2026
- Best Student Presentation Award, The 38th New England Statistics Symposium 2025

- Outstanding Teaching Assistant Award in IEOR, Columbia University 2025
- Second Place, 2023 INFORMS Blue Summit Supplies Data Challenge 2023
- Samuel N. Rubinstein Fellowship, Columbia University 2022
- Academic Performance Scholarship, CUHK(SZ) 2019–2022

Presentations

- Uncertainty Quantification for LLM-Based Survey Simulations
 - Talk: NYC Operations Day PhD Colloquium (March 2025), Columbia Foundations of Data Science Workshop (April 2025), New England Statistics Symposium (June 2025), INFORMS Annual Meeting (October 2025)
 - Poster: Columbia AI Summit (March 2025), NYC Operations Day (March 2025), Columbia Data Science Day (April 2025), Columbia Optimization and Statistical Learning Workshop (April 2025), Columbia Foundations of Data Science Workshop (April 2025), Cornell ORIE Young Researchers Workshop (October 2025)
- Model Assessment and Selection under Temporal Distribution Shift
 - Talk: New England Statistics Symposium (May 2024)
 - Poster: Columbia Data Science Day (April 2024), Columbia Foundations of Data Science Workshop (April 2024), New York Academy of Sciences Machine Learning Symposium (October 2024)
- A Stability Principle for Learning under Non-Stationarity
 - Talk: NYC Operations Day PhD Colloquium (May 2024), INFORMS Annual Meeting (October 2024)
 - Poster: NYC Operations Day (May 2024), International Conference on Continuous Optimization (July 2025)

Professional Service and Activities

- Reviewer for ICML and ICLR
- Co-organizer of NYC Operations Day PhD Student Colloquium
- Student host of Columbia IEOR-DRO Seminar
- Co-organizer of Columbia IEOR Colloquium

Teaching Experience

At Columbia University, as an Instructor:

- IEOR E4502 - Python for Analytics: Fall 2025

At Columbia University, as a Graduate Teaching Assistant:

- IEOR E8100 - High-Dimensional Probability with Applications: Spring 2024, Spring 2026
- IEOR E4101 - Probability, Statistics and Simulation: Fall 2024
- IEOR E3658 - Probability for Engineers: Spring 2025
- IEOR E3402 - Production Planning and Inventory Control: Spring 2023
- IEOR E3106 - Stochastic Systems and Applications: Fall 2023, Fall 2025

At The Chinese University of Hong Kong, Shenzhen, as an Undergraduate Student Teaching Fellow:

- MAT3006 - Real Analysis: Spring 2021, Fall 2021
- MAT2006 - Elementary Real Analysis: Fall 2020
- MAT1001 - Calculus: Fall 2019